

Algebra II with Trigonometry

Chapter 11.2

Mental Math (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:

[4666322_alg-2trig-h-chp-11-2-note-docx.pdf](#)

1. **Axis Check** For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$, is the major axis horizontal or vertical?
2. **Vertices Fast** For $\frac{x^2}{36} + \frac{y^2}{16} = 1$, what are the vertices?
3. **Minor Axis Length** For $\frac{x^2}{49} + \frac{y^2}{25} = 1$, what is the length of the minor axis?
4. **Find c** For $\frac{x^2}{25} + \frac{y^2}{16} = 1$, what is c ?
5. **Find the Foci** For $\frac{x^2}{64} + \frac{y^2}{36} = 1$, what are the foci?
6. **Eccentricity Quick** For $\frac{x^2}{100} + \frac{y^2}{64} = 1$, what is the eccentricity?
7. **Vertical Ellipse** For $\frac{x^2}{9} + \frac{y^2}{49} = 1$, what are the vertices and foci?
8. **Build the Equation** An ellipse has foci $(\pm 4, 0)$ and vertices $(\pm 5, 0)$. What is its standard equation?
9. **Circle or Ellipse?** If an ellipse has $a = b$, what shape is it, and what is its eccentricity?
10. **Intersection by Symmetry** The ellipses $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and $\frac{x^2}{9} + \frac{y^2}{16} = 1$ intersect at four points. What are those points?